

PALEONTOLOGY AND STRATIGRAPHY

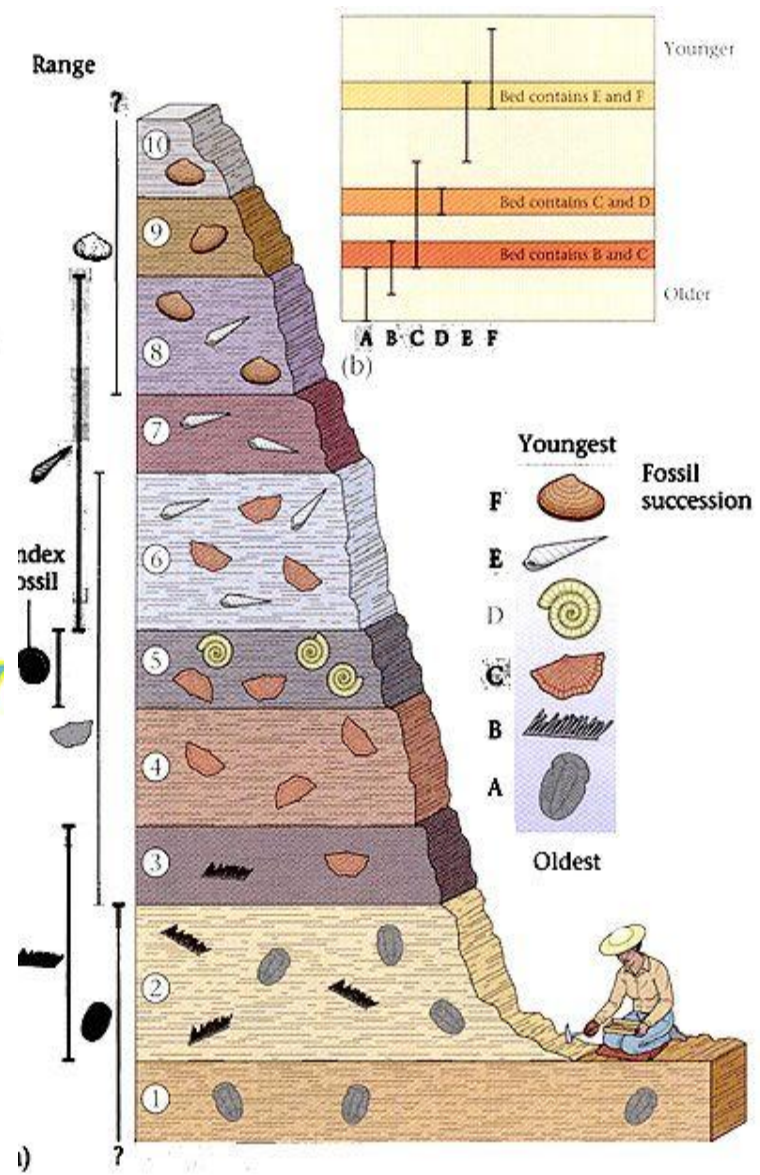
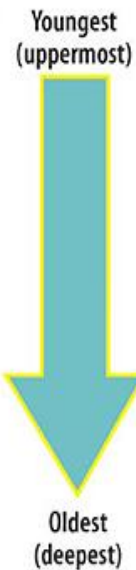
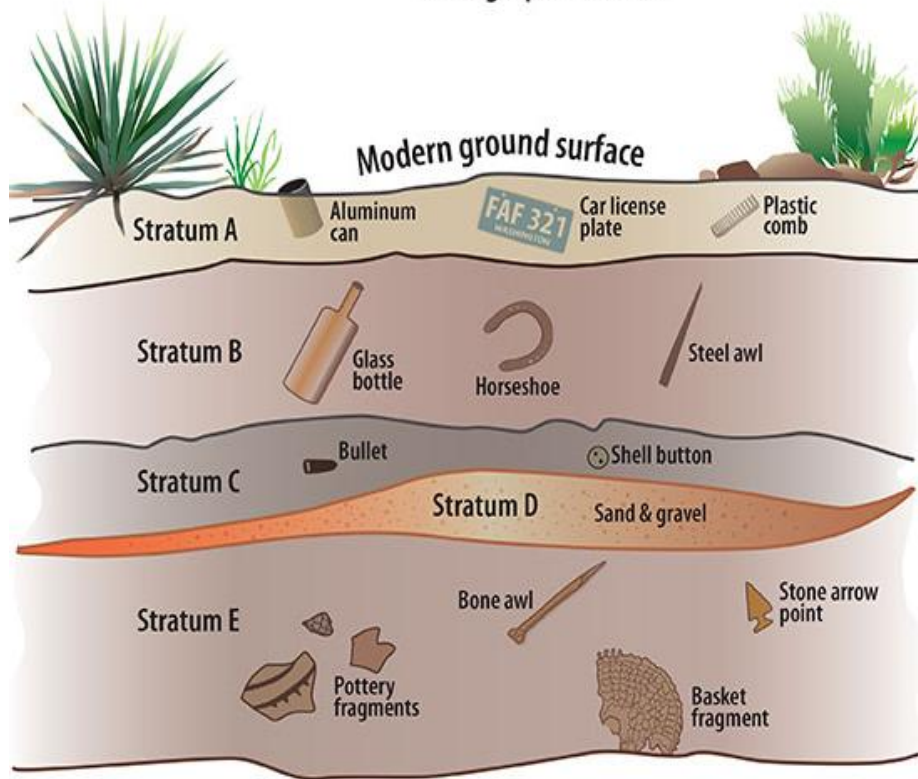
Paleontology is the study of the history of life on Earth as based on fossils.

Fossils are the remains of plants, animals, fungi, bacteria, and single-celled living things that have been replaced by rock material or impressions of organisms preserved in rock.

Stratigraphy is a branch of geology concerned with the study of rock layers (strata) and layering (stratification). It is primarily used in the study of sedimentary and layered volcanic rocks.



Stratigraphic Profile



Paleontology has various subdisciplines:

Micropaleontology: Study of generally microscopic fossils, regardless of the group to which they belong.

Paleobotany: Study of fossil plants; generally includes the study of fossil algae and fungi in addition to land plants.

Palynology: Study of pollen and spores, both living and fossil, produced by land plants.

Invertebrate Paleontology: Study of invertebrate animal fossils, such as mollusks, echinoderms, and others.

Vertebrate Paleontology: Study of vertebrate fossils, from primitive fishes to mammals.

Human Paleontology (Paleoanthropology): The study of prehistoric human and proto-human fossils.

Taphonomy: Study of the processes of decay, preservation, and the formation of fossils in general.

Ichnology: Study of fossil tracks, trails, and footprints.

Paleoecology: Study of the ecology and climate of the past, as revealed both by fossils and by other methods.

PRINCIPLES OF STRATIGRAPHY AND STRATIGRAPHIC CODE

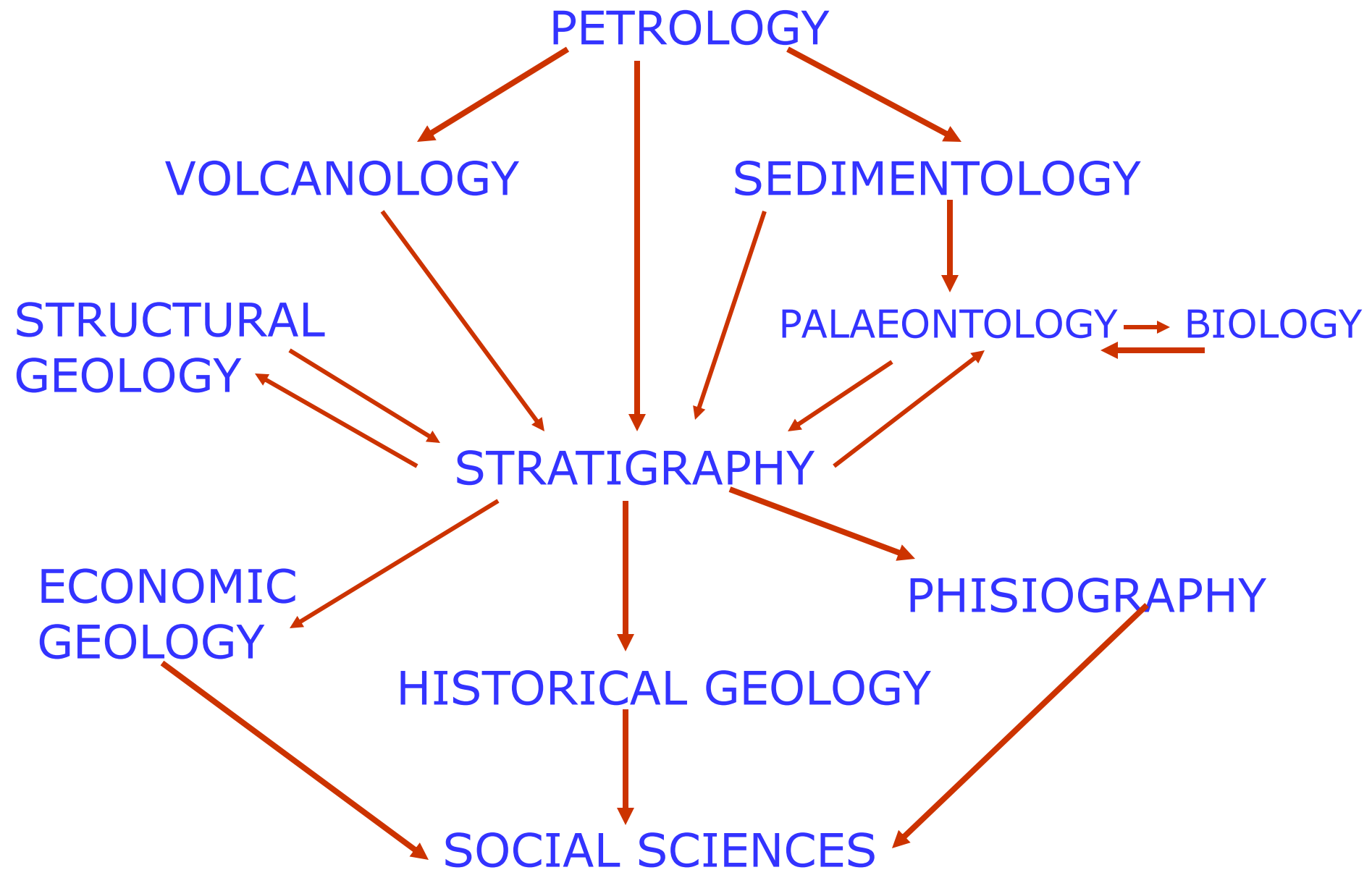
PRINCIPLES OF STRATIGRAPHY

Stratigraphy is based on a set of principles that govern the processes of sedimentation and concerns with the study of rock layers (strata). It involves

- Description and classification of rock strata.
- Observing the mode of distribution of rock strata
- Determination of spatial and temporal relationships of the strata
- Correlation of strata from an area with another
- Arranging them in a chronological order in which they were laid down on the surface of the earth.

Materials of Stratigraphy

- Sediments and rocks
- Good outcrops
- General lithological characters like sandstone siltstone, shale, limestone, coal, evaporates, bedded chert, sedimentary iron ore.
- Colour -grey, yellow, brown etc.
- Structural features like bedding, ripple marks concretions etc.
- Textural characteristics- size (clay, silt, sand, pebble, gravel, cobble, boulder), shape, roundness and sphericity etc.
- Deformational events



STRATIGRAPHY IS THE FOCAL POINT WHERE MOST GEOLOGIC KNOWLEDGE IS SYNTHESIZED INTO THE RECONSTRUCTION OF GEOLOGIC HISTORY

Stratigraphic laws include

Horizontality- the laws of gravity will deposit the sediments horizontal to the surface of the earth and the beds will pinch out along the sides of the basin.

Lateral continuity- Beds can be traced over a long distance if the basin is open type.

Law of inclusions- rock fragments in the stratum are older than the stratum itself

Order of Superposition-The strata are deposited in a sequence such that the deeper the layer, the older the rock and the younger are nearest the surface. The bottom layers are older than the top layers ie. Bottom layers were deposited first. This is the case only in the undisturbed sedimentary sequence.

- **Law of faunal** succession-The oldest fossils in a series of sedimentary rocks will be found in the lowest layer.

Similar fossils in different areas indicate time relations of geologic events.

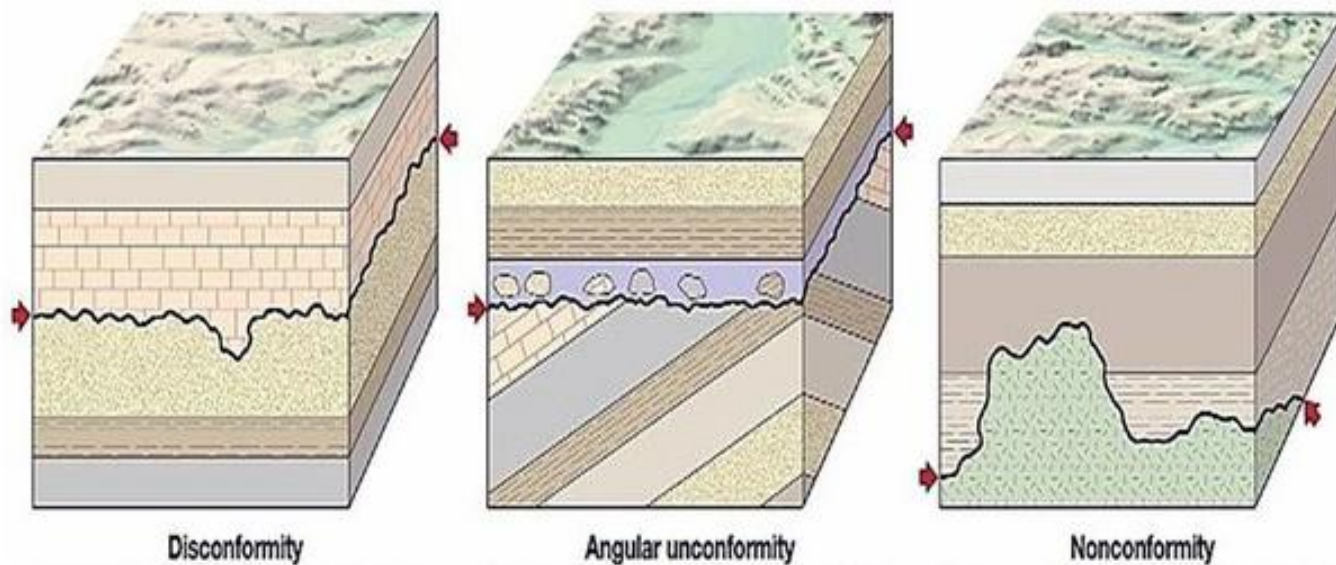
Unconformities-represents time gap between two stratigraphic layers.

Different types of stratified rocks occur in a definite sequence and they can be identified by the topographic expression, by the type of soils developed over them and the type of fossils they contain.

Unconformity

It is a stratigraphic plane that marks an important interruption in the stratigraphic record. It is the time interval that is unrepresented in the stratigraphic record.

An **unconformity** is a contact between two rock units in which the upper unit is usually much younger than the lower unit.



Angular unconformity: Angular unconformity older strata are tilted/deformed, eroded, and exposed as a depositional surface and an unconformable sequence of younger strata are deposited above them.

There is a difference in strike or dip or both between the two sets of strata.

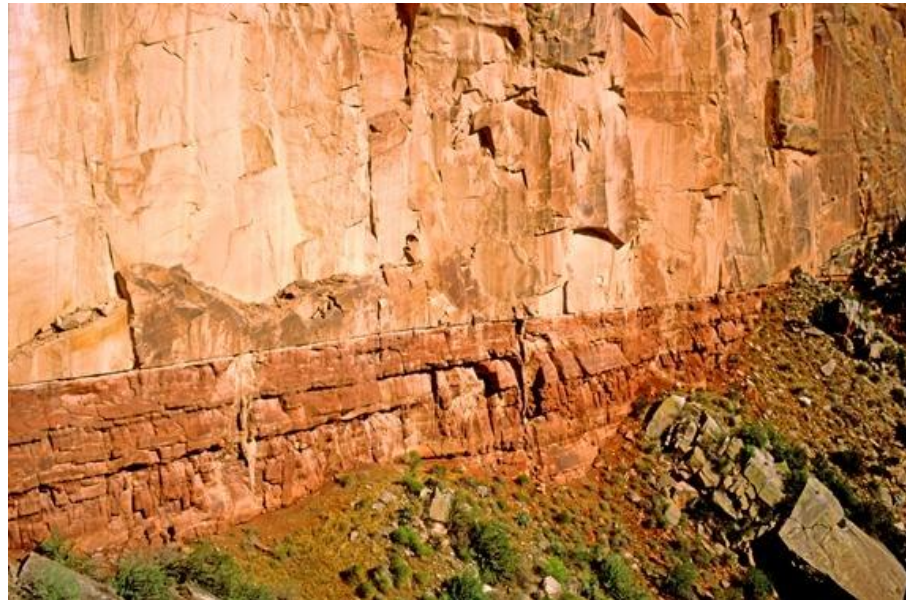


Disconformity (formations show parallelism)

Disconformity strata are eroded to yield a new depositional surface.

The unconformable overlying strata have the same strike and dip as the underlying strata.

This may be identified by different fossils or by examination of the actual contact in the field.



Nonconformity

Where a sedimentary formation overlies an igneous or metamorphic rock there is obviously a significant stratigraphic break.





Order of superposition

Definitions and procedures

- **Stratum:** A geologic stratum is a layer of rock characterised by certain, properties or attributes that distinguish it from adjacent layers
- **Stratigraphic unit:** A stratum or assemblage of adjacent strata recognised as a unit (distinct entity)
- **Stratigraphic classification:** Systematic **organisation** of the Earth's rock strata, as they are found in their original sequence, into units with **reference to any of the characters, properties, etc.**

Stratigraphic Categories

- **Lithostratigraphy:** organisation of strata into units based on their lithological character
- **Biostratigraphy:** organisation of strata into units based on their fossil content
- **Chronostratigraphy:** organisation of strata into units based on their age relationship

Lithostratigraphic units

Hierarchy:

Supergroup-a set of two or more associated groups

Group – two or more formations

Formation – Primary unit of lithostratigraphy

Member – Named lithologic unit within formation

Bed – Named unit or a layer in a member or formation

A group may on occasion be divided into subgroups.

The term supergroup may be used for several associated groups.

A complex is a lithostratigraphic unit composed of diverse types of any class or classes of rock and characterised by highly complicated structure

Boundaries of lithostratigraphic units are placed at positions of lithologic change. Selection of boundary may be by lateral traceability, physiographic expression, fossil content, lithogenesis and well-log character

-----Unconformity-----

Nallamalai Group (3500-6000)	Cumbum Formation	(2000)	Mainly slate and phyllite, chert, dolomite
	Pullampet Formation		Shale, felsic tuff, barytes, dolomite, quartzite
	Bairenkonda Quartzite (1500)/		Quartzite, quartz wacke, shale
	Nagari Quartzite (4000)		Quartzite, quartz wacke, shale, basal conglomerate

-----Angular unconformity-----

Chitravati Group (4900-5000)	Gandikota Quartzite (300)		Shale-quartzite alternations, glauconitic quartzite interbeds
	Tadpatri Formation (4600)		Shale, felsic tuff, basic sills and flows, chert, stromatolitic dolomite, quartzite
	Pulivendla Quartzite (1-75)		Quartzite and conglomerate
Papaghni Group (2100)	Vempalle Formation (1900) with Kuppalapalle Volcanics		Stromatolitic dolomite, chert, quartzite, basic sills and flows
	Gulcheru Quartzite (30-210)		Basal conglomerate, arkose, quartzite with shale interbeds

Crystalline basement of Dharwar craton

*Number in parenthesis refer to thickness in metres

Nandyal Shale (50-100)	Shale, partly calcareous
Koilkuntla Limestone (15-50)	Siliceous, shaly limestone with quartzite interbeds
Paniam Quartzite (10-35)	Massive and pinnacled quartzite, siliceous shale, basal pebble bed

-----Paraconformity-----

Auk Shale (10-15)	Laminated shale, mostly ochreous with siltstone interbeds
Narji Limestone (100-200)	Flaggy grey limestone, glauconitic sandstone interbeds
Banganapalle Quartzite (10-50)	Oligomict conglomerate, 'grit', quartzite, shale

-----Unconformity-----

(300)	Srisailem Quartzite (300)	Glauconitic ferruginous quartzite with shale
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-----Unconformity-----

Biostratigraphic units

- Rock strata are classified biostratigraphically by dividing them into units distinguished by differences **in their fossil content**
- Units are based on discrete particles in the rocks(fossil remains)
- Fossils usually constitute only a minor, disseminated, fractional part of a rock stratum
- Fossils may be the remains of the organism that lived in the area or brought only after death
- Under certain circumstances, rocks may contain fossils of younger age due to contamination, infiltration or leaking. Such introduced fossils should be distinguished from indigenous fossils in biostratigraphic zonation

Biostratigraphic zones (contd.)

- **Biozone** is a general term for any kind of biostratigraphic unit
- **Assemblage Zone**: That thickness of strata characterised by a distinctive natural assemblage of all forms
- **Range-Zone**: That thickness of strata representing the stratigraphic range of some selected element
- **Acme-Zone**: A group of strata based on the abundance or development of certain forms, regardless of either association or range
- **Interval-Zone**: The stratigraphic interval between two biohorizons

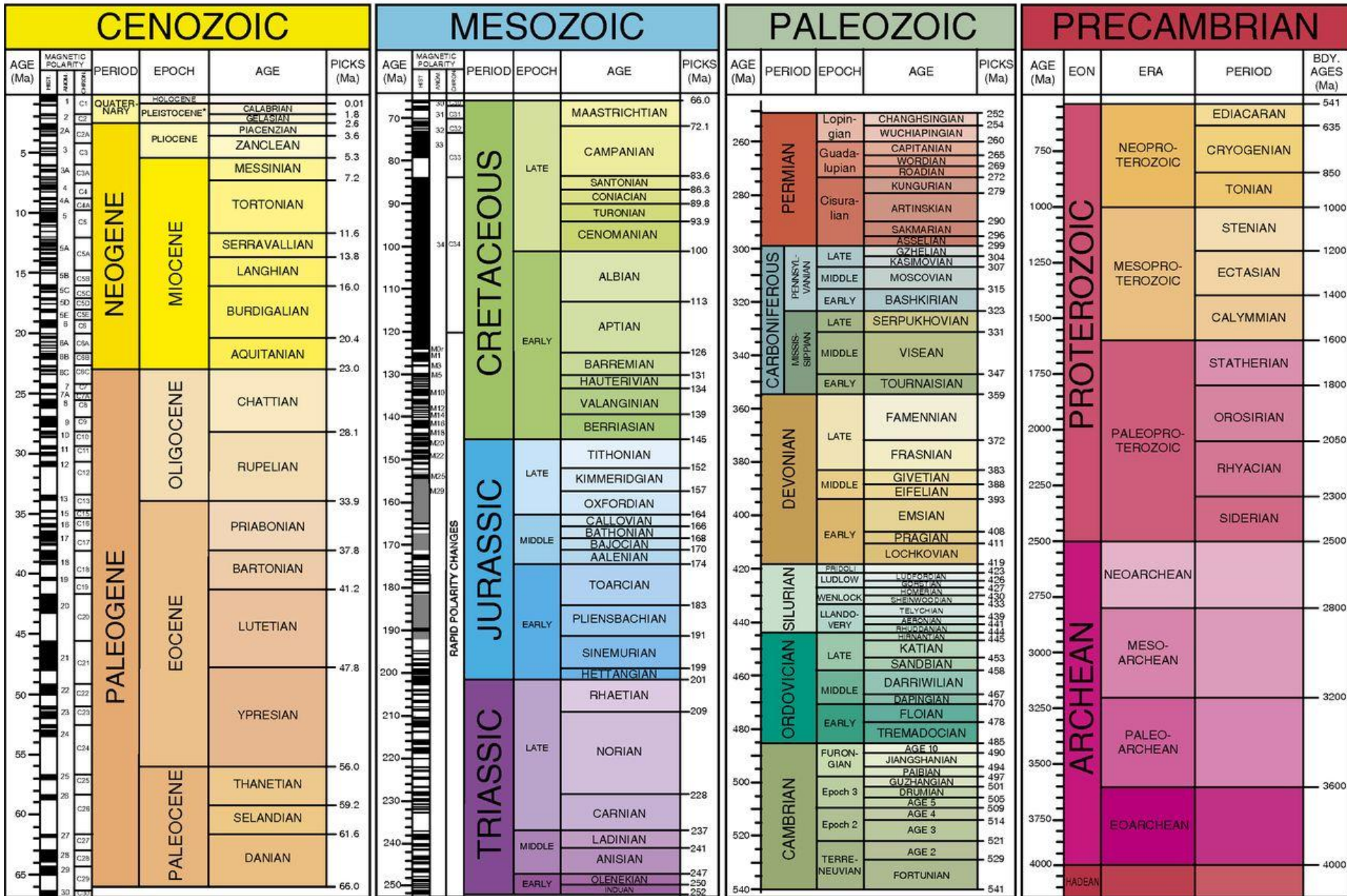
Geochronologic units (Geologic Time Units)

- Each interval of stratified rocks represents a certain interval of time.
- Geochronologic units are units of geologic time-an intangible unit- while stratigraphic units are tangible material units composed of rock strata.
- geochronologic units are not in themselves stratigraphic units.
- Each chronostratigraphic unit has a corresponding geochronologic unit.

Chronostratigraphic units and their corresponding geologic time units

Chronostratigraphic	Geochronologic
Eonothem	Eon
Erathem	Era
System	Period
Series	Epoch
Stage	Age
Chronozone	Chron

GEOLOGIC TIME SCALE



Era	Period	Epoch	Duration (millions of years)	Dates (millions of years)
Cenozoic	Quaternary	Recent	0.01	0.01
		Pleistocene	1	1
	Tertiary	Pliocene	10	11
		Miocene	14	25
		Oligocene	15	40
		Eocene	20	60
		Paleocene	10	70
Mesozoic	Cretaceous		65	135 ± 5
	Jurassic		30	165 ± 10
	Triassic		35	200 ± 20
Paleozoic	Permian		35	235 ± 30
	Pennsylvanian		30	265 ± 35
	Mississippian		35	300 ± 40
	Devonian		50	350 ± 40
	Silurian		40	380 ± 40
	Ordovician		70	460 ± 40
	Cambrian		90	550 ± 50
Precambrian			4500	5000

The image is a composite of four photographs of fossils. The top-left shows two ammonite fossils, one with a dark, patterned shell and another with a lighter, ribbed shell. The top-right shows a complete dinosaur skeleton, likely a T-Rex, standing on a light background. The bottom-left shows a fish fossil embedded in a light-colored rock matrix. The bottom-right shows a large trilobite fossil with distinct segments. The text 'ALL ABOUT' is in yellow, and 'Fossils' is in white, both centered over the top half of the image. 'Notes' is in grey, centered below the main title.

ALL ABOUT

Fossils

Notes

Requisites for Fossilization

Conditions favourable for preservation. We know that millions of animals and plants had lived, died and were destroyed without leaving a trace. But it has been observed that two factors are favourable for the preservation of organisms as fossils

- (a) the possession of hard parts such as shells and bones, and**
- (b) quick burials of the remains by different processes to prevent destruction by scavengers and decay.**

Fossils



- Fossils are preserved remains or traces of living things.
- Fossils normally form in sedimentary rock.
- Hard parts are the only parts of an organism that leaves a fossil.
- Examples: Bones, shells, teeth, seeds, and woody stems.

FIVE types of Fossils Found in Rock

are:

- 1. Molds
- 2. Casts
- 3. Petrified Fossils
- 4. Carbon Films
- 5. Trace Fossils



Environments for Fossil Formation

- Other fossils may be found in tar, amber, or ice.
- Large beds of sedimentary rocks indicate a former marine or aquatic/semi-aquatic environment
- Sedimentary rocks form when sediments (sand, silt, mud) are hardened into stone as the environment changed in response to changes in climate or geography.
- The presence of fossils in sedimentary rocks indicate that the past environment supported life.

Types of Fossils

- Molds

- A hollow area in sediment in the shape of an organism is a mold
- A mold forms when the hard part of the organism is buried in sediment.
- When water comes in and dissolves away the organism the hollow area left that is in the shape of the original organism is the mold.



Types of Fossils



REAL



CAST

- Casts
 - Water carrying dissolved minerals and sediments may seep into the mold.
 - This sediment may harden and take the shape of the mold making a copy of the fossil.
 - A cast is a solid copy of the shape of a fossil or organism.



Types of Fossils

- Casts
 - A cast is the opposite of its mold.
 - Both the mold and casts preserve details of the organism's structure.



Types of Fossils



- **Petrified Fossils**

- Petrified fossils are fossils in which minerals replace all or part of an organism
- Water rich in dissolved minerals seep into spaces of the organism.
- Over time, the minerals come out of the solution and harden filling in all the spaces. This causes the organism to be preserved.

Types of Fossils



- Carbon Films
 - Carbon films are an extremely thin coating of carbon on rock.
 - When sediment buries an organism, some of the materials that make up the organism evaporate or become gases.
 - These gases escape from the sediment, leaving carbon behind.
 - Eventually only a thin layer of carbon is left behind. This helps in preserving the organism.

Types of Fossils



- Trace Fossils
 - Trace fossils provide evidence of the activities of ancient organisms.
 - A fossilized footprint is a trace fossil.
 - From trace fossils, scientists can learn about a prehistoric organism's size, diet, environment, and behavior.

Perserved Remains

- Some processes preserve the remains of organism with little or no change.
- Processes that preserve the remains of organisms with little or not change include: Tar, Amber, and Ice



Application of Fossil Record

- All the information that paleontologists have gathered about past life is called the fossil record.
- The fossil record provides evidence about the history of life and past environments on Earth.
- It also shows how different groups of organisms have changed over time.
- The presence of coal in Antarctica shows that the climate there was once warmer.
- Scientists use the fossil record to support theories of how living things have changed over time.

Fossil Record

- The relative age of a rock is its age compared to other rocks
- The absolute age of rock is the number of years since that rock was formed.
- A fossil that is widely distributed and represents a type of organism that existed only briefly is called an index fossil.
 - They are useful because they can be used to identify the relative age of rocks.

Index Fossils

- Index fossils are used to define geological periods.
- These fossils can be defined as "commonly found, widely distributed fossils that are limited in time span."
- If one finds an index fossil in a given layer, then one has bounds on the age of the layer. Using index fossils, the geological periods are defined

Examples: Ammonites were common during the Mesozoic Era (245 to 65 mya), They were not found after the Cretaceous period, as they went extinct during the K-T extinction (65 mya).

Brachiopods (mollusk-like marine animals) appeared during the Cambrian (540 to 500 mya); some **examples** still survive.

Vertebraria India-Lower Gondwana

Ptiloophyllum Flora-Upper Gondwana